

Day 1.E.

ФУТ-231

Вопросы - 15

Задача 1

$$(e^x + 8)dy - ye^x dx = 0$$

$$(e^x + 8)dy = ye^x dx$$

$$\frac{dy}{y} = \frac{e^x dx}{e^x + 8} \Rightarrow \int \frac{dy}{y} = \int \frac{e^x dx}{e^x + 8}$$

$y \neq 0$. Проверка?

$$1) \int \frac{dy}{y} = \ln|y| + C$$

$$2) \int \frac{e^x dx}{e^x + 8} = \left| \begin{array}{l} e^x + 8 = t \\ t' = e^x \\ dt = e^x dx \end{array} \right| = \int \frac{dt}{t} = \ln|t| + C = \ln|e^x + 8| + C$$

$$\ln|y| = \ln|e^x + 8| + C \quad \text{o.f.n}$$

Задача 2a

$$y' = \frac{x^2 + 2xy - y^2}{2x^2 - 2xy}; \quad y' = f(x, y)$$

$$2x^2 \neq 2xy, \quad 2x(x-y) \neq 0, \quad \begin{pmatrix} x \neq 0 \\ x \neq y \end{pmatrix}$$

$$1) f(2x, 2y) = \frac{(2x)^2 + 2(2x)(2y) - (2y)^2}{2(2x)^2 - 2(2x)(2y)} = \frac{4x^2 + 8xy - 4y^2}{8x^2 - 8xy} = \frac{4(x^2 + 2xy - y^2)}{8(x^2 - xy)} = \frac{x^2 + 2xy - y^2}{2(x^2 - xy)} = f(x, y)$$

т.к. $f(2x, 2y) = f(x, y) \Rightarrow$ ДГом-е

$$2) y = tx \Rightarrow y' = t'x + t$$

$$t'x + t = \frac{x^2 + 2tx^2 - t^2x^2}{2x^2 - 2tx^2} \Rightarrow t'x = \frac{x^2(1 + 2t - t^2)}{x^2(2 - 2t)} - t \Rightarrow t'x = \frac{1 + 2t - t^2 - 2t + 2t^2}{2 - 2t} \Rightarrow$$

$$t'x = \frac{1 + t^2}{2 - 2t} \Rightarrow \frac{dt}{dx} x = \frac{1 + t^2}{2 - 2t} \Rightarrow \int \frac{2 - 2t}{1 + t^2} dt = \int \frac{dx}{x} \quad | x \neq 0$$

$$1) \int \frac{2 - 2t}{1 + t^2} dt = \int \frac{2}{1 + t^2} dt - \int \frac{2t}{1 + t^2} dt = 2 \arctg(t) - \int \frac{2t}{1 + t^2} dt = 2 \arctg(t) - \int \frac{du}{u} = 2 \arctg(t) - \ln|u| + C$$

$$= 2 \arctg(t) - \ln|1 + t^2| + C \Rightarrow 2 \arctg\left(\frac{y}{x}\right) - \ln\left|1 + \frac{y^2}{x^2}\right| + C$$

$$2) \int \frac{dx}{x} = \ln|x| + C$$

$$2 \arctg\left(\frac{y}{x}\right) - \ln\left|1 + \frac{y^2}{x^2}\right| = \ln|x| + C \quad \text{o.f.n}$$

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Задача 25

$$y' = \frac{y+3y-4}{5x-y-4} \quad \Delta = \begin{vmatrix} 1 & 3 \\ 5 & -1 \end{vmatrix} = -1-15 = -16 \neq 0 \Rightarrow \begin{cases} x=U+d \\ y=V+B \end{cases} \Rightarrow y' = \frac{dy}{dx} = \frac{dV}{dU}$$

$$\frac{dV}{dU} = \frac{U+d+3V+B-4}{5U+5d-V-B-4} = \frac{U+3V+(d+3B-4)}{5U-V+(5d-B-4)} \Rightarrow \begin{cases} d+3B-4=0 \\ 5d-B-4=0 \end{cases}$$

$$d=4-3B$$

$$20-15B-B-4=0$$

$$\begin{aligned} 16-16B=0 &\Rightarrow B=1 \\ d+3-4=0 &\Rightarrow d=1 \end{aligned} \Rightarrow \begin{cases} x=U+1 \\ y=V+1 \end{cases}$$

$$\frac{dV}{dU} = \frac{U+1+3V+3-4}{5U+5-V-1-4} = \frac{U+3V}{5U-V}$$

$$V' = \frac{U+3V}{5U-V} \quad V=tU \Rightarrow V' = t'U + t$$

$$t'U = \frac{U+3tU}{5U+tU} - t \Rightarrow t'U = \frac{1+3t-5+t^2}{5+t} \Rightarrow t'U = \frac{(t-1)^2}{5+t}$$

$$\frac{dt}{dU} \cdot U = \frac{(t-1)^2}{5+t} \Rightarrow \int \frac{5-t}{(t-1)^2} dt = \int \frac{dU}{U} \quad \left| \begin{array}{l} +71 \\ U \neq 0 \end{array} \right.$$

$$1) \int \frac{5-t}{(t-1)^2} dt = \left| \begin{array}{l} W=t-1 \\ W'=1 \\ dW=dt \end{array} \right| = \int \frac{5-W}{W^2} dW = \int \frac{5-(W+1)}{W^2} dW = \int \frac{4-W}{W^2} dW =$$

$$= \int \frac{4}{W^2} dW - \int \frac{W}{W^2} dW = -\frac{4}{W} - \ln|W| + C$$

$$1.1) \int \frac{4}{W^2} dW = 4 \int \frac{1}{W^2} dW = -\frac{4}{W} + C \quad 1.2) \int \frac{W}{W^2} dW = \ln|W| + C$$

$$2) \int \frac{dW}{W} = \ln|W| + C$$

$$-\frac{4}{t-1} - \ln|t-1| = \ln|U| + C$$

$$-\frac{4U}{V-U} - \ln\left|\frac{V}{U}-1\right| = \ln|U| + C$$

$$\left(-\frac{4x-4}{y-x} - \ln\left|\frac{y-1}{x-1}\right| - 1 \right) = \ln|x-1| + C$$

0,5-н Проверка особых решений?

(±)

Aufgabe 3a

$$\frac{dy}{dx} + \frac{2y}{x} = x^3; y(1) = -\frac{5}{6}$$

$$y' + \frac{2}{x}y = x^3; P(x) = \frac{2}{x}, Q(x) = x^3$$

$$y = U \cdot V$$

$$y' = U'V + UV'$$

$$U'V + U(V' + \frac{2}{x}V) = x^3$$

$$V' + \frac{2}{x}V = 0 \Rightarrow \frac{dV}{dx} + \frac{2}{x}V = 0 \Rightarrow \frac{dV}{dx} = -\frac{2}{x}V \Rightarrow \frac{dV}{V} = -\frac{2dx}{x}$$

$$\int \frac{dV}{V} = -2 \int \frac{dx}{x}$$

$$\ln|V| = -2 \ln|x| + C$$

$$e^{\ln|V|} = e^{\ln|x^{-2}|} \cdot e^C$$

$$V = \frac{1}{x^2}$$

$$U'V = x^3 \Rightarrow U' \cdot \frac{1}{x^2} = x^3 \Rightarrow U' = x^5 \Rightarrow U = \int x^5 dx = \frac{x^6}{6} + C$$

$$y = U \cdot V = \left(\frac{x^6}{6} + C\right) \cdot \frac{1}{x^2} = \frac{x^4}{6} + \frac{C}{x^2}$$

$$y(1) = \frac{1}{6} + C = -\frac{5}{6} \Rightarrow C = -1$$

$$y = \frac{x^4}{6} - \frac{1}{x^2}$$



Aufgabe 3b

$$3xy' + 5y = (4x-5)y'' \mid \cdot \frac{1}{3x} \quad x \neq 0$$

$$y' + \frac{5}{3x}y = \frac{4x-5}{3x}y'' \mid \cdot \frac{1}{y''} \quad y \neq 0$$

$$\frac{y'}{y''} + \frac{5}{3x} \cdot \frac{1}{y''} = \frac{4x-5}{3x}$$

$$z = \frac{1}{y''} \Rightarrow z' = -3y''y' = -3 \frac{y'}{y^4} \Rightarrow \frac{y'}{y^4} = -\frac{z'}{3}$$

$$\frac{z'}{-3} + \frac{5}{3x}z = \frac{4x-5}{3x} \mid \cdot -3$$

$$z' - \frac{5}{x}z = -4 + \frac{5}{x}$$

$$z = U \cdot V \Rightarrow z' = U'V + UV'$$

$$U'V + UV' = \frac{5}{x} \cdot UV = -4 + \frac{5}{x}$$

$$U'V + U(V' - \frac{5}{x}V) = -4 + \frac{5}{x}$$

$$V' - \frac{5}{x}V = 0$$

$$\int \frac{dV}{V} = \int \frac{5}{x} dx \Rightarrow \ln|V| = 5 \ln|x| + C$$

$$e^{\ln|V|} = e^{\ln|x^5|} + C$$

$$V = x^5 \Rightarrow U'x^5 = -4 + \frac{5}{x} \Rightarrow U' = -\frac{4}{x^5} + \frac{5}{x^6}$$

$$U = \frac{1}{x^4} - \frac{1}{x^5} + C$$

$$Z \cdot U \cdot V = \left(\frac{1}{x^4} - \frac{1}{x^5} + C \right) \cdot x^5 = x - 1 + Cx^5$$

$$x: Z \cdot \frac{1}{y^3} \Rightarrow y^3 = \frac{1}{Z} \Rightarrow y = \left(\frac{1}{Z} \right)^{\frac{1}{3}}$$

$$y = \left(\frac{1}{x - 1 + Cx^5} \right)^{\frac{1}{3}}$$

$$y(1) = \left(\frac{1}{C} \right)^{\frac{1}{3}} = 1 \Rightarrow C = 1$$

$$y = \sqrt[3]{\frac{1}{x^5 + x - 1}}$$

Проверка ограничений?



Задача 4

$$\underbrace{(1 + e^{\frac{x}{y}})}_P dx + \underbrace{e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right)}_Q dy = 0$$

$$\left. \begin{aligned} \frac{dP}{dy} &= \frac{d}{dy} (1 + e^{\frac{x}{y}}) = e^{\frac{x}{y}} \left(-\frac{x}{y^2} \right) = -\frac{x}{y^2} e^{\frac{x}{y}} \\ \frac{dQ}{dx} &= \frac{d}{dx} e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) = \frac{1}{y} e^{\frac{x}{y}} - \frac{x}{y^2} e^{\frac{x}{y}} = -\frac{x}{y^2} e^{\frac{x}{y}} \end{aligned} \right\} \text{ДуфрД}$$

$$\frac{dU}{dx} = 1 + e^{\frac{x}{y}}$$

$$\frac{dU}{dy} = e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right)$$

$$U = \int (1 + e^{\frac{x}{y}}) dx = x + y e^{\frac{x}{y}} + \varphi(y)$$

$$\frac{dU}{dy} = \frac{d}{dy} x + y e^{\frac{x}{y}} + \varphi(y) = e^{\frac{x}{y}} - \frac{x}{y} e^{\frac{x}{y}} + \varphi'(y) = e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) + \varphi'(y)$$

$$e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) + \varphi'(y) = e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) \Rightarrow \varphi'(y) = 0 \Rightarrow \varphi(y) = C_1 \Rightarrow U = x + y e^{\frac{x}{y}} + C_1$$

$$C_2 = x + y e^{\frac{x}{y}} + C_1$$

$$x + y e^{\frac{x}{y}} = C$$



Задача 5

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$$xy' - y = \ln y'$$

$$y = xy' + f(y')$$

$$y = xy' - \ln y'$$

$$p = y' \Rightarrow y = xp - \ln(p) \quad (p = p(x))$$

Диф-н по x:

$$dy = xdp + p dx - \frac{dp}{p}$$

$$\frac{d}{dx} y = \frac{d}{dx} (xp - \ln(p))$$

$$y' = (xp)' - \ln(p)'$$

$$p = x' \cdot p + x \cdot p' - \ln(p)' \cdot p'$$

$$p = p + x \frac{dp}{dx} - \frac{1}{p} \cdot \frac{dp}{dx} \Rightarrow$$

$$\Rightarrow x \frac{dp}{dx} - \frac{1}{p} \frac{dp}{dx} = 0$$

$$\frac{dp}{dx} \left(x - \frac{1}{p} \right) = 0 \Rightarrow$$

$$\begin{cases} dp = 0 \\ x - \frac{1}{p} = 0 \end{cases}$$

Рассм. 2 сл-я:

Общее решение

$$1) \frac{dp}{dx} = 0 \Rightarrow p = C \Rightarrow y = xC - \ln(C) \mid C > 0 - \text{д.р. (семейство прямых)}$$

$$2) x - \frac{1}{p} = 0 \Rightarrow p = \frac{1}{x} \Rightarrow y = 1 + \ln(x) - \text{особое решение (определяет семейство прямых)}$$

где g - то, что
это решение вообще?

⊕

T/p: g2